A Proof Dependency Manifest for a Candidate Computer-Assisted Proof of $f(7,7) = 38$

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Abstract

This document is a dependency manifest, not a paper. It states exactly what the candidate proof of $f(7,7) = 38$ consists of, and separates published external mathematics from project-original arguments, computational reductions, external database checks, certificate verification, Lean-kernel verification, and the items still requiring human review. Every claim carries a source and a verification status. It is written so that a referee can locate the weakest link quickly.

Status. This document describes a *candidate* computer-assisted proof. All currently known computational components have been independently checked or formally verified, but the interpretation of AKR Observation 6 / Lemma 7 remains subject to external human review. The result should not be treated as established.

Disclosure of generative-AI use

Generative-AI systems — principally Anthropic's Claude, together with OpenAI's ChatGPT — were used substantially throughout this project. Their use was not incidental: it extended to proof exploration and the search for the reduction strategy; literature-guided reasoning about the published results; generation of essentially all software, in Python, C++ and Lean; the computational experimentation itself; debugging; the construction of redundant and independent verification paths; the Lean formalisation; source auditing of the cited papers; and the drafting of the project's documentation, including this manifest.

The named author is responsible for the claims made in this document. AI-generated output was not treated as inherently trustworthy at any point. Each major claim is discharged by an explicit, separately recorded mechanism — source audit against the published papers, independent reimplementation in a second language, an externally sourced dataset, a machine-checkable certificate with a standalone checker, or Lean-kernel verification — and §3 records which mechanism applies to which claim. Several substantive errors introduced during AI-assisted development were found by those mechanisms and are catalogued in §11; they are retained there rather than quietly repaired, because they calibrate how much checking the argument has actually received.

Project artifacts and provenance — source code, generated data, certificates, build and verification logs, the audit documents, and the complete development transcripts — are preserved in the project repository.

No claim is made here about conformance with any particular publisher's or institution's policy on generative-AI use; that must be assessed separately against the policy in force.

1 Claim and current status

Let $f(m, n)$ denote the maximum number of boundary intersection points of two simple polygons with m and n sides in general position.

Theorem 1 (Claimed)**.** $f(7, 7) = 38$.

Three facts frame the problem.

1. 38 **is realizable.** An explicit pair of integer-coordinate heptagons attains 38 crossings (§9, LB38); this is verified by the Lean kernel.

2. **The published state of the art leaves only** 40**.** Černý et al. [5, Theorem 4] prove $f(k, \ell) \le k\ell - \lceil k/6 \rceil - \ell$ for odd $7 \le k \le \ell$, which at $k = \ell = 7$ gives $f(7, 7) \le 40$; [2] is asymptotically optimal but vacuous at $m = n = 7$. Crossings of two closed curves in general position are even, so $f(7, 7) \in \{38, 40\}$ and the whole question is whether 40 is attainable.

   This bound is context, not a dependency. Importing it would shorten the argument — $I \le 40$, $I \ne 40$ and parity would give $I \le 38$ at once, with no $I = 42$ case — but [5, Theorem 4] rests on [5, Lemma 3], a four-page case analysis this project has not audited, and importing it would contradict the project's own accounting of its dependencies. The argument here instead uses only [5, Lemma 1] (C1), which gives the weaker cap $I \le 42$, and excludes $I = 42$ separately in §4.3. Černý Theorem 4 is therefore independent corroboration of the upper bound, not an input to it.

3. **The upper bound is a proof by contradiction.** One assumes a pair with $I = 40$ and derives a contradiction. Everything in §2–§8 serves that single reductio.

We do *not* claim Theorem 1 is established. We claim that no computational gap is currently known, and that the remaining risk is concentrated in one identified place (A7-EXT, §4.6).

2 One-page proof flow

Write G^D for the bipartite *disjointness graph* on the $7+7$ sides, with $(p_i, q_j) \in E^D$ iff the closed segments p_i and q_j are disjoint. Since two non-collinear segments meet at most once, crossings biject with intersecting side pairs, so

$$I + |E^D| = 49.$$

The reductio proceeds as follows; the label under each arrow names the justification.

$I = 40$	$\xrightarrow{I + \lvert E^D \rvert = 49}$	$\lvert E^D \rvert = 9$
	$\xrightarrow{\text{C1 (Černý L1)}}$	$\delta(G^D) \ge 1$
	$\xrightarrow{\text{LA (Lemma A + parity)}}$	P and Q are both *exceptional*
	$\xrightarrow{\text{OT,EX (order-type classification)}}$	$(\text{profile}_P, \text{profile}_Q) \in \{14\} \times \{14\}$
	$\xrightarrow{\text{GR,COMP (enumeration; A5-C,A7-EXT,C2)}}$	225 terminal triples on 121 graphs
	$\xrightarrow{\text{A5-A,CERT,LEAN-END}}$	every terminal triple violates AKR Lemma 5
	$\Longrightarrow$	$\bot$.

Hence $I \ne 40$; with $I \le 42$ (C1), $I \ne 42$ (§4.3) and I even, this gives $I \le 38$, and the explicit construction LB38 attains 38. Therefore $f(7, 7) = 38$.

Remark. All of this is conditional on the general-position regime GP, imported from [2, §9.1]; see §4.1.

3 Dependency table

ID	Claim	Type	Source	Verification	Remaining risk
GP	General-position regime may be assumed w.l.o.g.	published theorem	[2, §9.1,§3]	SOURCE-AUDITED	none known; project's own earlier proof was wrong and is removed
C1	A line meets a simple 7-gon in at most 6 sides; hence $\delta(G^D) \geq 1$	published theorem	[5, Lemma 1]	SOURCE-AUDITED	none known
C2	Forbidden local matching on 4 consecutive sides with $D \in \operatorname{int} \triangle ABC$	published theorem	[5, Lemma 2]	SOURCE-AUDITED	none known
C2-BRIDGE	The implemented filter predicate fires iff an *exact induced* perfect matching exists	project inference	this project	LEAN-KERNEL-CHECKED	certifies the Lean transcription of the filter, not the Python/C++ source
A5-C	Lemma 5 applied to *consecutive* side pairs in different components	published theorem	[2, Lemma 5]	SOURCE-AUDITED	none known
A5-A	Lemma 5 applied to *avoiding* side pairs	published theorem	[2, Lemma 5]	SOURCE-AUDITED; bridge LEAN-KERNEL-CHECKED	none known
A7-EXT	Under the intended reading of "associated"/"is hooking", AKR Lemma 7 gives: adjacent H-vertices have disjoint status sets	published theorem + interpretation	[2, Lemma 7], [1]	VERY STRONGLY CORROBORATED	**principal open item**; Observation 6 is unproved and *false as printed* (§11, item 6), though the failure cannot reach $S(e)$; four bridges from the machine checks to the geometry remain prose
A7-BRIDGE	Non-empty status sets + disjoint status sets across edges $\Rightarrow$ H bipartite	project inference	this project	LEAN-KERNEL-CHECKED	none known; conditional on A7-EXT
LA	Lemma A: an uncrossable 4-set forces $\lvert E^D \rvert \geq 10$, parity gives ≥ 11, so $I \leq 38$	project lemma	this project	LEAN-KERNEL-CHECKED (counting core)	geometric premises h_1, h_2 remain hypotheses
OT	There are exactly 135 order types of 7 points	external dataset cross-check	[4, 3]	EXTERNALLY CROSS-CHECKED (exact set equality)	none known
EX	6,342 simple heptagons; 16 exceptional; 14 profile classes	finite enumeration	this project	INDEPENDENTLY REIMPLEMENTED; 6,342 EXTERNALLY CROSS-CHECKED	none known

ID	Claim	Type	Source	Verification	Remaining risk
GR	Exactly 13,003,620 candidate disjointness graphs	finite enumeration	this project	INDEPENDENTLY REIMPLEMENTED + closed form	none known
COMP	Filter pipeline down to 121 graphs / 225 triples	finite enumeration	this project	INDEPENDENTLY REIMPLEMENTED (exact set match)	inherits A7-EXT
CERT	Each of the 225 triples violates the AKR-L5 consequence	certificate	this project	CERTIFICATE-CHECKED	completeness of the list is not certified
LEAN-LB	The explicit pair has 38 crossings and is simple	Lean formalization	`lean/LowerBound.lean`	LEAN-KERNEL-CHECKED	simplicity is orientation-level, not Jordan-curve
LEAN-LA	Counting and parity core of Lemma A	Lean formalization	`lean/LemmaA.lean`	LEAN-KERNEL-CHECKED	proved as an implication
LEAN-END	All 225 cases fail the AKR-L5 condition; conclusion $\leftrightarrow$ predicate	Lean formalization	`lean/Endgame.lean`	LEAN-KERNEL-CHECKED	completeness of the 225 list not formalized
LB38	$f(7,7) \geq 38$	construction	this project	LEAN-KERNEL-CHECKED	none known
UB42	$I = 42$ is impossible	project inference	this project, §4.3	MANUALLY CHECKED; exhaustively swept	consumes C1, LA, EX only — *not* A7-EXT; C1 gives only $I \leq 42$, so this branch is needed
UB40	$I = 40$ is impossible, hence $f(7,7) \leq 38$	composite	GP, C1, C2, A5-C, A5-A, A7-EXT, A7-BRIDGE, LA, COMP, UB42	conditional on A7-EXT	**fails if A7-EXT fails**

4 External mathematical dependencies

4.1 GP — general position, AKR §9

Role. The whole combinatorial argument is carried out in a general-position regime. GP is what licenses that restriction.

Source. [2, §9.1, and §3]. Their perturbation removes the relevant degeneracies without decreasing the number of boundary intersections.

Exact consequences used, and nothing stronger.

(GP1) both polygons remain simple;

(GP2) no vertex of one polygon lies on the boundary of the other, so every intersection point is interior to two sides — this is what allows intersections to be counted as proper crossings;

(GP3) no three vertices of P and Q combined are collinear;

(GP4) no two sides of P and Q combined are parallel;

(GP5) consequently the supporting line $\ell(q_j)$ of any side of Q contains no vertex of P (it passes through two vertices of Q, so a vertex of P on it would violate (GP3)). This is exactly the hypothesis that Lemma A needs, and it is the *only* kind of line at which Lemma A is ever instantiated.

Remark (A superseded project argument). Earlier versions of this project proved GP by a project-specific perturbation rule. That argument is **incorrect**: its rule for a vertex-on-vertex degeneracy, "push the coincident vertex into $\mathrm{Cone}_Q(w)$", destroys the intersection point whenever $\mathrm{Cone}_P(v) \subset \mathrm{Cone}_Q(w)$, yielding 0 crossings instead of ≥ 1. It also never addressed the case in which one side participates in several degeneracies at once. It has been removed from the active proof and replaced by the citation above. See §11.

Remark (Counting model). [2, §9.3] explicitly leaves the variant counting intersecting *side pairs* rather than intersection *points* as future research. Everything here is in the intersection-*point* model, and the result must not be restated in the pairs model.

4.2 C1 — Černý Lemma 1

A line meets a simple k-gon an even number of times, hence at most $k-1$ times for odd k [5, Lemma 1]. Applied to $\ell(p_i)$ against Q and to $\ell(q_j)$ against P, every side misses at least one side of the other polygon. The single consequence used is

$$\delta(G^D) \;\geq\; 1.$$

Together with $|E^D| = 49 - I$ this also gives $I \leq 42$.

4.3 Excluding $I = 42$

The manifest previously asserted twice that $I = 42$ is "excluded separately" without giving the argument. It is this. (This subsection is a *project-original* argument — dependency UB42 — placed here because it consumes C1 directly; it is not itself an external result.)

$I = 42$ forces $|E^D| = 7$. With $\delta(G^D) \geq 1$ on $7 + 7$ vertices and only 7 edges, every vertex has degree exactly 1, so G^D is a *perfect matching*. Its components are therefore the 7 matched pairs: each component contains exactly one side of P and one of Q. In particular no two sides of P share a component.

Next, both polygons are *exceptional*. Lemma 2 with $|S| = 4$ would force $|E^D| \geq 7 + (7-4) = 10 > 7$, so neither P nor Q admits an uncrossable 4-set. Hence the crossing-set data of §5.2 applies to each.

Now fix j. Since G^D is a perfect matching, $N(q_j) = \{i_j\}$ is a singleton, so the set of sides of P that q_j *crosses* is $\{0, \ldots, 6\} \setminus \{i_j\}$, of size 6. Compatibility requires this crossing set to be contained in a maximal crossing set of P's class. Every exceptional class has exactly five maximal crossing sets, each of size 6 and of the form $\{0, \ldots, 6\} \setminus \{m\}$ for $m \in M_P$, with $|M_P| = 5$. Two 6-subsets of a 7-set nest only if they are equal, so $i_j \in M_P$. This holds for every j, and the i_j are distinct because G^D is a matching, so all seven sides of P lie in M_P — contradicting $|M_P| = 5$. Hence $I \neq 42$.

Remark. Verified by exhaustive sweep over all $7! = 5{,}040$ perfect matchings against all 14 profile classes: 0 survive. Re-run against the independent profile reconstruction of §7 (whose class labels differ from the shipped ones by a dihedral relabelling of side indices, matched 14/14): also 0. The two invariants consumed — $|M| = 5$, and maximal crossing sets $= \{\textsc{full} \setminus \{m\} : m \in M\}$ — were recomputed from the class data rather than read from the stored invariant block, and hold for all 14 classes in both reconstructions.

Remark (This branch does not depend on A7-EXT). Earlier versions of this manifest excluded $I = 42$ through A7-EXT: a perfect matching makes every consecutive pair relevant and every index-non-adjacent pair span four distinct components, so H is the complement of the 7-cycle, which contains the triangle $e_0e_2e_4$ and is not bipartite. That argument is correct but unnecessary. The route above consumes only C1, LA and the exceptional-class invariants EX, all of which are elementary and order-type based, so no further external lemma is imported — in

particular not [5, Lemma 3], whose four-page case analysis this project has not audited and which an earlier draft wrongly treated as the only available reroute.

The practical consequence is that A7-EXT is load-bearing at exactly *one* point in the proof: the $I = 40$ endgame. It cannot be dropped there — with the filter off the pipeline ends at 2,187 surviving triples on 412 graphs, not 0 (§6).

Remark. The shipped $|E^D| = 7$ sweep reproduces the older route independently: 5,040 → 14 (A5-C) → 0 (A7-EXT). The companion $|E^D| = 8$ sweep gives 476,280 → 1,078 → 0, for $I = 41$, which parity already excludes. Both were a by-product of the historical enumeration bug of Remark 6.1 and are corroborative rather than load-bearing.

4.4 C2 — Černý Lemma 2

The only consequence used is a local forbidden configuration. Let A, B, C, D, E be five consecutive vertices of P, giving four consecutive sides AB, BC, CD, DE, and suppose $D \in \operatorname{int} \triangle ABC$. Then for no four sides of Q is the induced subgraph of G^D on those four sides of P and those four sides of Q *exactly* a perfect matching [5, Lemma 2].

The geometric hypothesis depends only on the order type and is tabulated per exceptional class as the *Lemma-2 windows*. The word *exactly* is load-bearing: a block containing a perfect matching *plus* an extra edge is not forbidden, and the implementation was checked not to reject such blocks.

4.5 A5 — AKR Lemma 5

Statement used. If two sides p_a, p_b of P that are *consecutive or avoiding* lie in different components of G^D, then there are consecutive sides $q_i, q_{i\pm1}$ of Q with

$$(p_a, q_i),\ (p_b, q_{i\pm1}) \in E^D \qquad \text{and} \qquad (p_a, q_{i\pm1}),\ (p_b, q_i) \notin E^D.$$

A5-C, consecutive use. Applied to every consecutive pair in different components; this is the first filter of the pipeline.

A5-A, avoiding use. Two non-adjacent sides are *avoiding* when neither contains the intersection point of their supporting lines. This is an order-type predicate, tabulated per exceptional class. A5-A is the final filter and the one the certificate records.

Combinatorial consequence. Writing $N(\cdot)$ for neighbourhoods in G^D, the displayed conclusion is equivalent to

$$\exists\, k \in N(p_a) \setminus N(p_b), \quad \exists\, k' \in N(p_b) \setminus N(p_a), \qquad k' \equiv k \pm 1 \pmod 7. \tag{1}$$

Both directions are immediate by unfolding: $k \in N(p_a) \setminus N(p_b)$ says exactly $(p_a, q_k) \in E^D$ and $(p_b, q_k) \notin E^D$.

Remark. This equivalence is **Lean-kernel-checked**, as `akr5Conclusion_iff_adjacent` in `lean/EndgameDefs.lean`, and depends on no axioms. It matters because it shows the implemented predicate is neither a weakening nor a strengthening of the published conclusion.

Remark. If $\mathrm{CC}(p_a) \neq \mathrm{CC}(p_b)$ then $N(p_a) \cap N(p_b) = \emptyset$, since a shared neighbour would join them in one component. The set differences in (1) are then the neighbourhoods themselves.

4.6 A7-EXT and A7-BRIDGE — AKR Lemma 7 and bipartiteness of H

This dependency was previously recorded as a single item. It bundles two things of very different character, and is split here:

$$\underbrace{\text{A7-EXT}}_{\text{published theorem+interpretation}} \implies \underbrace{\text{A7-BRIDGE}}_{\text{project inference, Lean-checked}} \implies H \text{ bipartite.}$$

Definition of H. Fix P (symmetrically for Q). Vertices are the relevant consecutive pairs, $V(H) = \{\, i : \mathrm{CC}(p_i) \neq \mathrm{CC}(p_{i+1}) \,\}$, and $i \sim_H k$ iff the two pairs share no side, i.e. $(k-i) \bmod 7 \notin \{0,1,6\}$, *and* the four sides $p_i, p_{i+1}, p_k, p_{k+1}$ lie in four distinct components of G^D. Condition (b) is imposed directly, so an H-edge reproduces Lemma 7's hypothesis exactly.

For $e \in V(H)$ let $A(e)$ be the set of pairs of consecutive sides of Q associated to e *in the narrow sense* — carrying, beyond the two edges of E^D, the *cone dichotomy* of Lemma 5 — and put

$$S(e) = \{\text{hooking} : \exists \alpha \in A(e),\ I(\alpha) \in \mathrm{Cone}(e)\} \cup \{\text{hooked} : \exists \alpha \in A(e),\ I(e) \in \mathrm{Cone}(\alpha)\}.$$

$S(e) \neq \emptyset$: $e \in V(H)$ is Lemma 5's hypothesis, so $A(e) \neq \emptyset$, and Lemma 5's dichotomy realises at least one status.

Remark (Which extra conditions the narrow sense actually needs). Lemma 5 delivers, beyond the two edges $(p_i, q_j), (p_{i+1}, q_{j\pm1}) \in E^D$, both the two cross non-edges and the cone dichotomy. Only the second is a genuine strengthening.

The cross non-edges are automatic. If $(p_i, q_{j\pm1}) \in E^D$ then $p_i - q_{j\pm1} - p_{i+1}$ is a path in G^D, so $\mathrm{CC}(p_i) = \mathrm{CC}(p_{i+1})$, contradicting $e \in V(H)$. This is pure connectivity; bipartiteness of G^D plays no part. Checked exhaustively over all 2^{16} bipartite graphs on $4+4$ vertices: 110,304 qualifying configurations, 0 violations.

The cone dichotomy is not. Adding the non-edges does *not* recover Observation 6: for $a = [(0,0),(2,0)]$, $b = [(0,0),(0,3)]$, $A = [(-1,10),(\frac{1}{10},\frac{6}{5})]$, $B = [(-1,10),(\frac{7}{5},-2)]$ all four edge and non-edge conditions hold with a, b consecutive, yet the axis $y = -10x$ leaves both a and b in $y \geq -10x$.

So the narrow notion used here is **edges + cone dichotomy**, the non-edges being free. A formalisation that instead defined it as "edges + cross non-edges" and then asserted Observation 6 would be unsound.

A7-EXT — external, very strongly corroborated

> Under the intended interpretation of "associated" and "is hooking", [2, Lemma 7] yields: if $e \sim_H e'$ then $S(e) \cap S(e') = \emptyset$.

This is the one item that remains subject to external human review, for three reasons, in increasing order of comfort.

1. AKR's Observation 6 (the Axis Property) is stated *without proof* in both versions, and is *false as printed*. The counterexample is not a bare four-segment picture but a pair of genuine simple polygons in full general position,

   $$P = (2,0),(3,4),(-2,0); \qquad Q = (4,-4),(-3,1),(0,-1),(3,3),(2,1),$$

   verified in exact integer arithmetic (§11, item 6). Here q_0, q_4 is associated to p_0, p_1 under the printed definition — indeed the cross non-edges hold too — yet the axis separates nothing. Observation 6 is used throughout Lemma 7's case analysis.

   This does *not* refute A7-EXT. Each status is by definition one disjunct of the cone dichotomy, so an associated pair failing the dichotomy contributes nothing to $S(e)$ and can never participate in the hypothesis. Equivalently, Lemma 7's hypothesis supplies the dichotomy exactly on the pairs that matter (Remark 4.6).

2. The conference version [1] opens its proof of Lemma 7 by *selecting* the associated pairs subject to the cone conditions $I(q_{i'}, q_{i'\pm1}) \in \mathrm{Cone}(p_i, p_{i+1})$, which forces the existential reading of "is hooking" used above. The journal version deleted those clauses, leaving its own next sentence unsupported. This strongly indicates the intended reading but does not state it in one place.

3. [2, §10.2], journal only, reports the authors' own independent computer verification of Lemma 7: 158,830 arrangements of eight lines, 840 assignments each after symmetry, yielding 14 configurations of which 13 fail Lemma 5's hooking/hooked pattern. This corroborates the *statement*. It also corroborates the *interpretation*: the stated ground for discarding those 13 is that "some associated pair did not conform to the hooking/hooked pattern of Lemma 5", a sentence that is coherent only if "associated" already carries the dichotomy.

   Two of the four published numbers in §10.2 do not reproduce, neither affecting the conclusion. 158,830 is a *nine*-element count (abstract order types on 9 points); x-monotone PSLAs of 8 pseudolines correspond to abstract order types on 9 points, the ninth element being the line at infinity, so a complete but redundant superset was enumerated — harmless for an impossibility claim. $840 = 8!/48$ quotients by Aut of the disjointness graph alone ($\cong Q_3$, order 48), a group that does not preserve which pairs are consecutive; the stabiliser of the configuration has order 16, giving $8!/16 = 2520$, reproduced by brute force over S_8. Both are fair questions for the authors.

Independent verification of A7-EXT

An oriented-matroid (rank-3 chirotope) encoding over the twelve segment endpoints makes the both-hooking and both-hooked patterns UNSAT, while the mixed patterns of AKR's Figure 6 are SAT. Unsatisfiability over a *superset* of realisable configurations is sound for an impossibility claim, so under-constraining is safe and only over-constraining would be fatal. The encoding was attacked on exactly that point:

- every encoded predicate was checked against the paper, and each constraint is a *necessary* consequence of the hypothesis;
- **positive control:** AKR's own Figure 6 configuration, reconstructed from integer line coefficients in exact rationals, satisfies all 64,566 clauses — so the encoding is not vacuously over-constrained;
- an independent re-encoding from scratch reproduces all 20 verdicts;
- re-solved with CaDiCaL, Glucose and MiniSat, all agreeing;
- 15 DRAT proofs verified by a from-scratch RUP/RAT checker, itself validated on PHP(4) and PHP(5);
- the chirotope axiom set validated by reproducing the published enumeration counts $1, 1, 1, 4$ for $n = 3, \ldots, 6$ (OEIS A006248);
- coverage recomputed independently: the generic, P-adj, Q-adj and PQ-adj cases exhaust $n = 7$, *including* the cyclic-distance-2 cases AKR's own described filter excludes.

Separately, a direct search for a counterexample on real heptagon pairs over exact integers, with simplicity and full general position enforced, examined ∼57,000 admissible configurations and 306 associated pairs on consecutive pairs in different components: 0 violations.

Remark (Why this is not VERIFIED INDEPENDENTLY). The machine checks are certificates about a chirotope encoding. Four steps connecting them to the geometric statement are hand-checked prose and none is certified: (i) general position of the ≤ 12 vertices is WLOG (inherited from GP; perturbation can change G^D); (ii) "four distinct components" yields exactly the encoded 20-pair pattern; (iii) the four cases above exhaust $n = 7$; (iv) Lemma 7's quantification matches the existential $S(e)$. The translation of "segments cross" and "point in cone" into chirotope conditions is validated by exact random testing, not by proof. And $S(e) \neq \emptyset$, needed alongside A7-EXT, rests on Lemma 5, which is machine-checked nowhere.

A7-BRIDGE — project inference, lean-kernel-checked

> If every relevant vertex has a non-empty status set, and adjacent vertices have disjoint status sets, then H is bipartite.

Proof. Let $e \sim_H e'$. Both $S(e), S(e')$ are non-empty and disjoint subsets of a two-element set, hence distinct singletons. So if $S(e) = \{\text{hooking}, \text{hooked}\}$ then e has no H-neighbour at all; every non-isolated vertex has $|S(e)| = 1$; every H-edge joins opposite singletons; and $e \mapsto S(e)$ is a proper 2-colouring of the non-isolated part, isolated vertices being irrelevant to 2-colourability. □

This inference is formalised in `lean/Lemma7Bridge.lean` (§9). The disjointness condition is an explicit *hypothesis*; AKR Lemma 7 is not assumed, not axiomatised, and not named in the file.

Remark (What the Lean check does and does not buy). It settles the "different associated pairs witness different statuses" worry completely: a both-status vertex is *proved* isolated, so the scenario in which Lemma 7 is applied with inconsistent choices on different H-edges is vacuous. It says nothing whatever about whether AKR's published language licenses A7-EXT. Lean verifies the project's implication; it does not verify AKR's geometric theorem.

5 Project-original mathematical reductions

5.1 Lemma A

Lemma 2 (Lemma A)**.** *Let $S \subseteq \text{sides}(P)$ satisfy*

$$(\star) \qquad \forall \ell, \quad \ell \cap V(P) = \emptyset \implies \ell \text{ misses at least one member of } S.$$

Then $|E^D| \geq n + (m - |S|)$.

Proof sketch. For each side q_j of Q, (GP5) gives $\ell(q_j) \cap V(P) = \emptyset$, so $(\star)$ applies and yields $p \in S$ with $\ell(q_j) \cap p = \emptyset$, hence $q_j \cap p = \emptyset$. This is n edges with distinct Q-endpoints, all with P-endpoint in S. By C1 each $p \notin S$ misses some side of Q, giving $m - |S|$ further edges with distinct P-endpoints outside S. The two families are disjoint because their P-endpoints lie inside versus outside S. □

For $m = n = 7$ and $|S| = 4$ this gives $|E^D| \geq 10$; since I is even, $|E^D| = 49 - I$ is odd, so $|E^D| \geq 11$ and $I \leq 38$. Call P *exceptional* when no 4-subset of its sides satisfies $(\star)$. Then

$$I = 40 \implies P \text{ and } Q \text{ are both exceptional.}$$

Lean boundary. `lean/LemmaA.lean` kernel-checks the counting and the parity upgrade (`card_E_ge`, `card_E_ge_eleven`, `crossings_le_38`) as an *implication*. The geometric premises — that each q_j is disjoint from some member of S, and that each $p \notin S$ is disjoint from some q_j — remain explicit hypotheses, discharged in Lean only for the one explicit certificate pair. The $(\star)$ condition itself is not formalized.

5.2 Exceptional classification

All predicates used — simplicity, crossing sets, the avoiding relation, the Lemma-2 windows — are Boolean functions of the order type of the seven vertices. The classification is therefore finite:

$$\begin{aligned} 135 \text{ order types} \times 360 \text{ cyclic orders} &\longrightarrow 6{,}342 \text{ simple heptagons} \\ &\longrightarrow 16 \text{ exceptional} \longrightarrow 14 \text{ classes.} \end{aligned}$$

The 6,342 split as 6,007 with an uncrossable 3-set, 319 with an uncrossable 4-set, and 16 exceptional. Each of the 14 classes has $|M| = 5$, exactly five maximal crossing sets all of size 6, and two avoiding pairs — except one class with none.

Remark (Strict versus loose, and why it matters). Two readings of "uncrossable" must be distinguished.

strict: ($\star$) above versus **loose:** no line at all meets all four closed sides.

Measured over all 6,342 simple heptagons these give

strict $\rightsquigarrow$ 16 exceptional, **loose** $\rightsquigarrow$ 2,268 exceptional,

differing on 2,252 objects. The *strict* reading is the correct one: Lemma A is only ever instantiated at lines $\ell(q_j)$, which avoid $V(P)$ by (GP5). The 2,252 differing objects are exactly those 4-sets whose only transversals pass through a vertex of P — and no $\ell(q_j)$ can be such a line. Had the loose phrasing been implemented, the finite search would have been infeasible.

6 Computational reduction

6.1 The universe of candidate graphs

There are $7 \times 7 = 49$ side pairs, and $I + |E^D| = 49$, so $I = 40$ forces $|E^D| = 9$. By C1 the minimum degree is at least 1 on both sides. The number of bipartite graphs on $7 + 7$ vertices with exactly 9 edges and minimum degree at least 1 is

$$\sum_{j=0}^{7} (-1)^j \binom{7}{j} [x^9]\big((1+x)^{7-j} - 1\big)^7 \;=\; 13{,}003{,}620,$$

by inclusion–exclusion over uncovered Q-sides. This closed form was evaluated in exact integer arithmetic independently of both enumerations.

Remark (A historical enumeration bug). An earlier implementation reported 13,484,940 as "the number of 9-edge graphs". Its recursion never forced the edge budget to be exhausted at the leaf, so it enumerated $|E^D| \in \{7, 8, 9\}$:

$$5{,}040 \;+\; 476{,}280 \;+\; 13{,}003{,}620 \;=\; 13{,}484{,}940.$$

The searched universe was therefore a *superset* of what was needed. No candidate was missed and the exclusion was not invalidated — only the reported figure was wrong, and the $|E^D| = 7$ and $|E^D| = 8$ cases were swept for free. The bug nonetheless motivated the independent reimplementation of §7.

6.2 The filter pipeline

Count	Filter applied next	Justification	Checked by
13,003,620	AKR L5, consecutive pairs	A5-C	Python, C++
42,238	AKR L7 $\Rightarrow H$ bipartite	A7-EXT + A7-BRIDGE	Python, C++
24,990	crossing-set compatibility	elementary (order type)	Python, C++
2,300	Černý L2, class-dependent	C2	Python, C++
121	*(change of counted object)*	—	Python, C++
225	AKR L5, avoiding pairs	A5-A	Python, C++, certificate, Lean
0	—	—	—

Remark (The 121 $\rightarrow$ 225 step is not a filter). This is a change in what is being counted, and it should not be read as a reduction. The figures 13,003,620 through 121 count *graphs*; the figure 225 counts *triples* $(\text{class}_P, \text{class}_Q, G)$. The 225 triples live on exactly those same 121 graphs: a graph may be compatible with several exceptional classes on either side, and $225 = \sum_G |ok_P(G)| \cdot |ok_Q(G)|$ over the 121 surviving graphs. Every one of the 121 graphs has degree sequence $(2, 2, 1, 1, 1, 1, 1)$ on both sides, and only 9 of the 14 classes occur.

Remark (A7-EXT cannot be dropped from this branch). The filter at 42,238 is the only place A7-EXT enters the proof (Remark 4.3), which raises the question of whether it is needed at all. It is. Re-running the whole pipeline with that filter switched off gives

$$13{,}003{,}620 \to 42{,}238 \to 6{,}376 \to 533 \to 2{,}412 \to \mathbf{2{,}187}$$

surviving triples, on 412 graphs — not 0. The structure is sharp: the 533 graphs surviving Černý L2 without A7-EXT split as exactly 121 H-bipartite + 412 non-bipartite; the 121 are identical to the shipped certificate's graphs; and the avoiding-pair filter A5-A kills all 121 and *none* of the 412. On the residual the two filters are perfectly complementary, so no reordering helps. No already-trusted ingredient substitutes: compatibility is already applied at full strength, and Černý Lemma 3 kills 0 of the 412.

This is evidence about the pipeline, not about AKR Lemma 7: it shows the present toolkit cannot avoid A7-EXT, not that A7-EXT is mathematically indispensable.

Remark (Černý Lemma 3 is not a dependency). An earlier pipeline also applied Černý Lemma 3 between A7-EXT and the compatibility filter, reducing 24,990 to 9,702. Running the pipeline with and without it produces *identical* results at every subsequent stage and an identical set of 225 terminal triples: the graphs Lemma 3 removes are all removed by crossing-set compatibility in any case. Lemma 3 is therefore *not* a logical dependency of this proof and is omitted throughout.

7 Independent implementations

Python (reference). The readable implementation: recursion over P-side neighbourhoods with a running edge budget, union-find components, DFS 2-colouring for bipartiteness, a 128-entry lookup table for compatibility, and iteration over all $\binom{7}{4}$ subsets for Černý L2.

C++17 (independent reduction). Written from a written specification, with every major algorithm chosen differently: enumeration transposed to Q-side columns and partitioned by an explicit degree sequence rather than an edge budget; graphs keyed as a packed 49-bit word; components by iterative bitmask closure; bipartiteness by union-find *with parity*; compatibility by direct containment in the five maximal crossing sets; Černý L2 by enumeration of systems of distinct representatives.

Agreement. Both produce the same universe 13,003,620; the same intermediate counts 42,238, 24,990, 2,300, 121; the same 121 graphs; and the same canonical set of 225 terminal triples, with *zero set difference* in either direction.

Remark (Scope of the independence claim). This is **implementation-independent in algorithm and data representation, but not fully author- or specification-independent**: the second implementation was written by an agent that had already seen the first implementation and its specification. The agreement is therefore strong evidence against transcription and coding errors, and against a missed branch in either enumeration, but it is *not* evidence against a shared misreading of the published lemmas.

8 Certificates

A *certificate* here is a file that lets a referee re-derive the terminal contradictions without trusting — or rerunning — the exhaustive search. For each of the 225 terminal triples it stores the class identifiers, a canonical graph encoding (seven neighbourhood bitmasks), the degree sequences, the component partitions, the relevant side pair and its kind, both neighbourhoods, the two set differences, and the list of cyclic adjacencies found, which is empty in every case.

The standalone checker recomputes every quantity from the canonical encoding: neighbourhoods, degree sequences, components (by breadth-first search, where the generator used

union-find), the set differences, and the adjacency test. The certificate's convenience fields are treated as claims to be checked, never as input.

Remark (Component labels are compared modulo relabelling). Component identifiers are arbitrary. The checker compares the induced *partitions*, not the integer labels. An earlier version of the comparison compared raw labels and reported spurious failures on 217 of 225 cases; the check was corrected to be relabelling-invariant.

Adversarial controls. The checker was validated against ten deliberate corruptions, all detected: a case marked as surviving, a pair not avoiding for its class, falsified neighbourhoods, falsified difference sets, a falsified adjacency list, a graph edited to break the edge count or minimum degree, a wrong degree sequence, a wrong component partition, and — the essential control — an *honestly constructed* witness using a different avoiding pair of the same class for which Lemma 5 genuinely *is* satisfied. That last case is rejected with a message naming the satisfying adjacency, which confirms the checker does not simply reject everything.

A worked terminal case

Take the first certificate entry. The graph is given by the P-side neighbourhoods

$$N(p_0), \ldots, N(p_6) \;=\; \{0\}, \{0\}, \{1\}, \{2,4\}, \{3\}, \{4\}, \{5,6\},$$

with P assigned exceptional class 12, whose M-set is $\{0,2,3,4,6\}$ and whose avoiding pairs are $(0,2)$ and $(4,6)$. The sides p_4 and p_6 are avoiding and lie in different components. Now

$$N(p_4) = \{3\}, \qquad N(p_6) = \{5,6\},$$

so $N(p_4) \setminus N(p_6) = \{3\}$ and $N(p_6) \setminus N(p_4) = \{5,6\}$. In $\mathbb{Z}_7$ the element 3 is cyclically adjacent only to 2 and 4, and neither lies in $\{5,6\}$. So (1) fails, contradicting A5-A. All 225 cases have this shape.

For contrast, the *other* avoiding pair $(0,2)$ of the same class on the same graph gives $N(p_0) \setminus N(p_2) = \{0\}$ and $N(p_2) \setminus N(p_0) = \{1\}$, and 0 *is* adjacent to 1 — so (1) holds there. The condition is genuinely discriminating: across the 225 certificate graphs, 2,016 of the 4,725 side pairs satisfy it.

9 Lean trust boundary

Environment: Lean 4.33.1; Mathlib tag `v4.33.1`, revision `0df444a3` [6]. Build: `lake build`.

None of the three files contains `sorry`, `admit`, a project-specific `axiom`, `unsafe`, or `native_decide`; this was verified by scanning the sources with comments stripped. All `decide` calls are ordinary kernel reduction.

`LowerBound.lean` — LEAN-LB

Imports: none. Core Lean only; verified to compile with an empty `LEAN_PATH`, so Mathlib is not in its trust base. **Axioms:** `propext` only.

Twenty theorems, including `P_simple`, `Q_simple` (distinct vertices; no three of the seven collinear; consecutive sides non-collinear; no proper crossing of non-adjacent sides), `general_position` (no vertex of one polygon on a side-line of the other), `crossing_count_eq_38`, `per_P_side_counts` $= [6,6,6,6,6,6,2]$, `disjoint_count_eq_11` and `crossCount_add_disjointCount`. Crucially `allPairs_length` $= 49$, `allPairs_nodup` and `allPairs_complete` establish that the count of 38 is taken over provably all 49 side pairs.

Not formalized: the step from these orientation-level facts to "the closed polygonal curve is a simple closed curve in $\mathbb{R}^2$".

`LemmaA.lean` — LEAN-LA

Imports: Mathlib (`Finset.Card`, `Fintype.Prod`, `Fintype.Card`, `Algebra.Ring.Parity`) and `LowerBound`. **Axioms:** `propext`, `Classical.choice`, `Quot.sound`.

Theorems `card_E_ge`, `card_E_ge_ten`, `ge_eleven_of_ge_ten_of_odd`, `card_E_ge_eleven`, `crossings_le_38`, together with a non-vacuity block (`Rcert_h1`, `Rcert_h2`, `Rcert_card_E` $= 11$, `nonvacuous`, `cert_I_le_38`, `hypotheses_are_restrictive`) showing the hypotheses are simultaneously satisfiable, the bound tight, and the empty relation excluded.

Remark (A defect in the earlier skeleton). An earlier, never-compiled version of this file did not have `Odd` in scope. Lean's auto-bound implicits silently turned it into a free variable, so `card_E_ge_eleven` and `crossings_le_38` elaborated as vacuous statements *with no error reported*. The file now sets `autoImplicit false`, which makes this class of failure impossible.

`Endgame.lean` — LEAN-END

Imports: `EndgameData` only, which imports `EndgameDefs`, which imports nothing — core Lean throughout, confirmed standalone with bare `lean` and `LEAN_PATH=`. **Axioms:** `propext` and `Quot.sound` only; notably *no* `Classical.choice`.

Principal theorems:

- `card_certificates` : the list has length 225;
- `keys_distinct` : no duplicate records under the canonical key;
- `all_225_cases_fail_AKR5_requirement` and `all_225_cases_fail_AKR5_conclusion`;
- `all_225_pairs_in_different_components` : component separation is *recomputed in the kernel* via a separator argument, not taken from the stored data;
- `akr5Conclusion_iff_adjacent` : the published conclusion and the implemented predicate are equivalent; this depends on no axioms at all;
- `akr5_not_vacuous` : 2,016 of the 4,725 side pairs in these graphs *do* satisfy the condition.

The Lean data file is generated deterministically from the committed certificate JSON, whose hash is embedded in the generated header.

`Lemma7Bridge.lean` — A7-BRIDGE

Imports: none. Core Lean; builds standalone with an empty `LEAN_PATH`. **Axioms: none at all** — every theorem reports *does not depend on any axioms*, not even `propext`.

Formalises only the implication of §4.6. The two hypotheses are `hne` : $\forall e, \exists s,\ S\,e\,s$ and `hdisj` : $\forall e\,e'\,s,\ \mathrm{Adj}\,e\,e' \to S\,e\,s \to S\,e'\,s \to \bot$. Theorems: `adj_irrefl` (irreflexivity is derived, not assumed), `isolated_of_both`, `status_unique_of_adj`, `statusIs_of_adj`, `opposite_of_adj`, `disjoint_statuses_imply_bipartite` (main), and a partition-form variant carrying no decidability hypothesis. Non-vacuity is proved rather than asserted: a concrete model with a real edge; a both-status vertex shown isolated; and a triangle shown both non-bipartite and admitting no valid status assignment, so the hypotheses genuinely constrain the graph.

AKR Lemma 7 does not appear — not as an axiom, not by name. Grepping the code with comments stripped finds no occurrence of AKR or Observation.

`CernyL2Interface.lean` — C2-BRIDGE

Imports: none. Axioms: the three standard ones; the pure `decide` facts are `propext`-only.

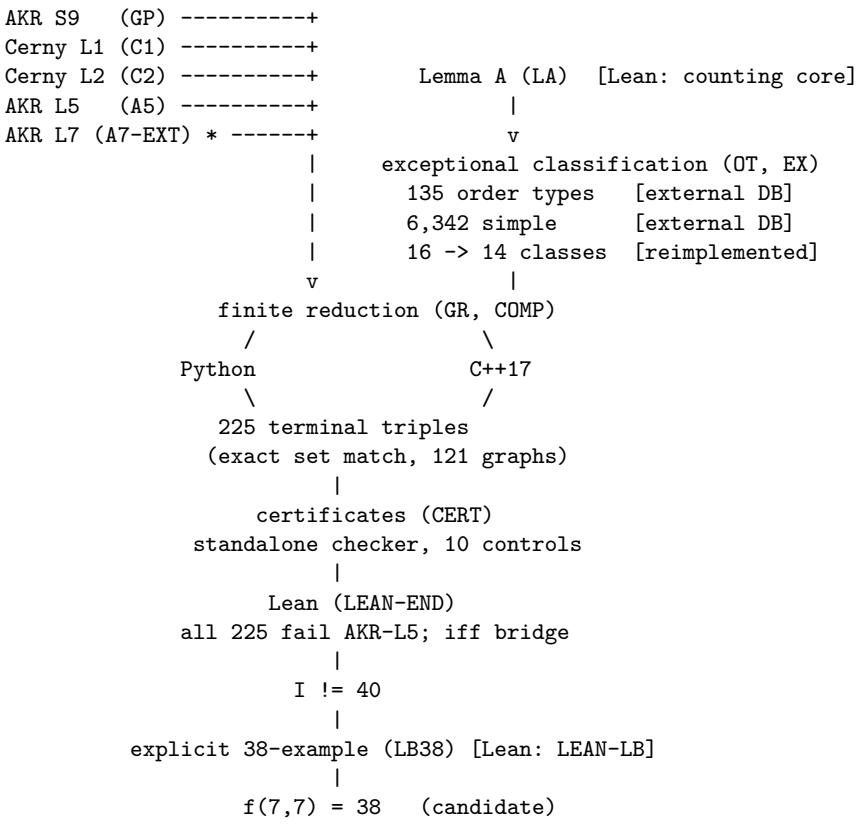
Proves `l2FiresB_iff`: the implemented filter fires iff an *exact induced* perfect matching exists, the single clause $\mathrm{Edge}\,(s_t)\,(q_u) \leftrightarrow t = u$ carrying both halves — all four matching edges present, all twelve cross pairs absent. The decisive fixtures: a matching plus an extra edge

inside the 4-block does **not** fire, while a matching plus an edge *outside* the block **still** fires, so the predicate is certified to be about the induced subgraph rather than about degrees.

Remark (Limits of C2-BRIDGE). It certifies the Lean transcription of the filter, *not* the Python/C++ source: there is no extraction, and the correspondence remains a human-eyeball link. One representational shortcut is unverified — the code computes popcount$(N_s \wedge S)$ where the Lean definition sums four indicator bits; these agree exactly when the four indices are distinct, which the enumeration enforces, but the bit-level identity is not proved. And it says nothing about Černý Lemma 2 itself, nor about the per-class Lemma-2 window sets, which remain fully external.

> Key finite and combinatorial components are formally verified in Lean; the published geometric lemmas themselves remain external dependencies. In particular $f(7,7) = 38$ is **not** a Lean theorem, and the completeness of the 225-case list is not formalized.

10 Trust DAG



* A7-EXT: VERY STRONGLY CORROBORATED, HUMAN REVIEW STILL DESIRABLE -- see the A7-EXT subsection of Section 4. The chain is

A7-EXT ==> A7-BRIDGE ==> H bipartite

where A7-EXT is the external reading of AKR Observation 6 / Lemma 7 (adjacent H-vertices have disjoint status sets), and A7-BRIDGE is the project's own inference from that to bipartiteness -- the latter is LEAN-KERNEL-CHECKED, with the disjointness condition as an explicit hypothesis and no AKR axiom.

If A7-EXT fails, the H-bipartite filter is unjustified and the reduction from 42,238 onwards must be redone; UB40 fails.

A7-EXT enters at exactly one point, the I=40 endgame: the I != 42 branch is routed around it through crossing-set compatibility alone, and the filter cannot simply be dropped (without it 2,187 triples survive).

11 Known defects found during development

Listed because they calibrate how much checking this argument has had.

1. **Enumeration bug (13,484,940).** The leaf test omitted the edge-budget condition, so $|E^D| \in \{7, 8, 9\}$ was enumerated. *Effect on the theorem:* none — a superset was searched, so no candidate was missed. *Repair:* corrected leaf test; the buggy file is preserved unpatched as evidence. *Superseded by:* the corrected enumeration and the closed form of §6.

2. **Raw component-ID comparison.** A verification script compared component *labels* rather than the partitions they induce, and reported failures on 217 of 225 cases. *Effect:* none on the proof; it was a checker bug. *Repair:* compare partitions. *Superseded by:* the current relabelling-invariant checker.

3. **Lean `Odd` / `autoImplicit` defect.** In the earlier `LemmaA.lean` skeleton, `Odd` was out of scope and became a free implicit variable, so two theorems elaborated as vacuous statements with no error. *Effect:* none, because the file had never been compiled. *Repair:* rewritten with `autoImplicit false`. *Superseded by:* the current `LemmaA.lean`.

4. **Step 0 perturbation argument.** The project's own general-position proof is incorrect when $\mathrm{Cone}_P(v) \subset \mathrm{Cone}_Q(w)$, where its rule destroys the intersection point. *Effect on the theorem:* none, because the required statement follows from [2, §9.1]. *Repair:* removed from the active proof. *Superseded by:* GP, §4.1.

5. **Strict versus loose exceptional condition.** Early phrasing said "no line meets all four sides", which does not license what the computation tests. *Effect if it had been implemented:* severe — 2,268 exceptional objects instead of 16, and an infeasible search. *Repair:* the strict, vertex-avoiding form ($\star$). *Superseded by:* Lemma 2 and Remark 5.2.

6. **akr Observation 6 is false as printed** — a defect in a published paper, not in this project. Stated without proof in both versions and used throughout the case analysis of Lemma 7. The counterexample is a pair of genuine simple polygons,

$$P = (2,0), (3,4), (-2,0); \qquad Q = (4,-4), (-3,1), (0,-1), (3,3), (2,1),$$

   verified in exact integer arithmetic and independently re-verified. Both are simple; the eight vertices are distinct; no three of the eight are collinear; no vertex of one lies on the other; no two of the eight sides are parallel; $I(P, Q) = 8$ is even — every general-position condition of [2, §3] holds. $\mathrm{CC}(p_0) \neq \mathrm{CC}(p_1)$ and $(p_0, q_0), (p_1, q_4) \in E^D$, so q_0, q_4 *is* associated to p_0, p_1 under the printed definition, and both cross non-edges hold as well; yet the cone dichotomy fails in both directions and the axis through the two apexes leaves a, b, A, B in one closed half-plane. *Effect on this proof:* none. A pair failing the dichotomy is neither hooking nor hooked, so it contributes nothing to $S(e)$ (Remark 4.6). *Effect on the manifest:* its earlier gloss of the narrow notion as "edges + cross non-edges" was wrong, since those non-edges are automatic and the dichotomy is what carries the content; corrected in §4.6.

12 Remaining work

1. External human review of the AKR Observation 6 / Lemma 7 interpretation (§4.6), and in particular of the four prose bridges of Remark 4.6.
2. Ideally, confirmation from one of the AKR authors: that "associated", as used in Observation 6 and Lemma 7, carries the cone dichotomy; and, separately, notice of the Observation 6 counterexample and of the two §10.2 counts that do not reproduce.
3. Independent expert review of the overall proof.

4. Conversion of this manifest into a readable paper.
5. The human author learning the argument well enough to defend it.
6. Further Lean formalization *only* where it improves the trust boundary.

Formalizing all of Černý et al. or all of AKR is **not** required by this proof architecture. Those results are used as published black boxes, and the appropriate check on them is source auditing and expert review, not formalization.

13 Final trust statement

The project currently contains a heavily audited *candidate* computer-assisted proof of $f(7,7) = 38$. The novel finite reductions and the terminal elimination have been independently implemented, certificate checked, and in key places Lean-kernel verified. The principal remaining mathematical issue is A7-EXT: whether the published Observation 6 / Lemma 7 language licenses precisely the disjoint-status consequence the argument uses. That consequence has since been independently corroborated — by proof-checked unsatisfiability of an oriented-matroid encoding with a verified positive control, and by exact-arithmetic search — but its four bridges to the geometric statement remain prose, so it is recorded as VERY STRONGLY CORROBORATED, not verified. The same investigation found AKR's Observation 6 to be false as printed, a defect that does not reach this argument (§11, item 6). Conditional on A7-EXT, the deduction that the auxiliary graph H is bipartite (A7-BRIDGE) is formally verified in Lean. Independent expert review remains required before the result should be treated as established.

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